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Internship : Data Analytics using Python
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Task 121 : Create data visualizations according to defined guidelines/policies.

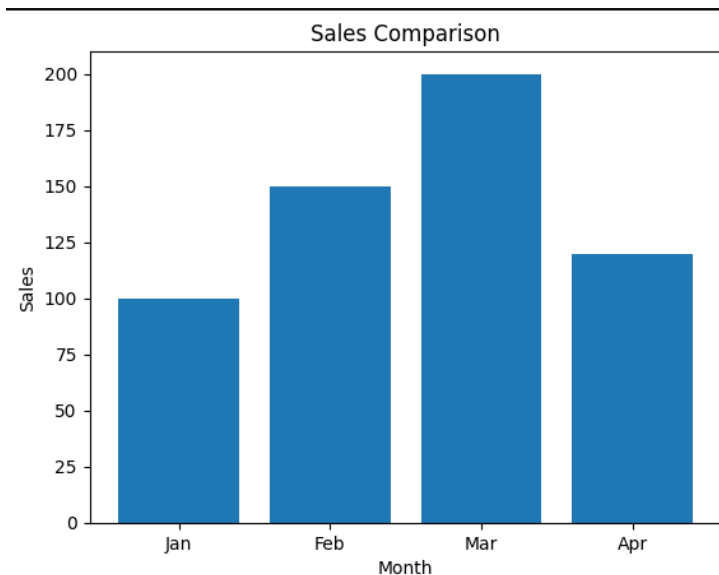
Objective

The objective is to design clear, accurate, and consistent data visualizations that simplify complex data and support better decision-making.

```
# 121. Create data visualizations according to defined guidelines/policies.
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr'],
    'Sales': [100, 150, 200, 120]
}

df = pd.DataFrame(data)
df
plt.figure()
plt.bar(df['Month'], df['Sales'])
plt.title("Sales Comparison")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Output :



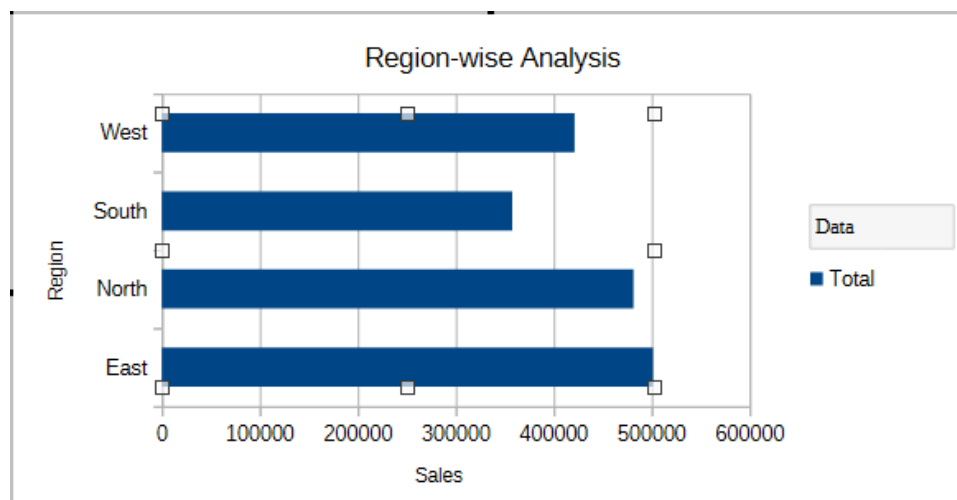
I created a data visualization by first organizing the dataset using a DataFrame and then plotting a bar chart to compare monthly sales. I followed visualization guidelines by adding a clear title, properly labeling the x-axis (Month) and y-axis (Sales), and choosing an appropriate chart type for comparison. This ensured that the data was presented accurately, consistently, and in an easy-to-understand format, helping users quickly interpret the sales trends.

Task 122 : Use different types of graphs to create an impactful report.

Objective

The objective of using different types of graphs in a report is to present data in a clear, meaningful, and visually engaging way that helps users quickly understand patterns, trends, and comparisons.

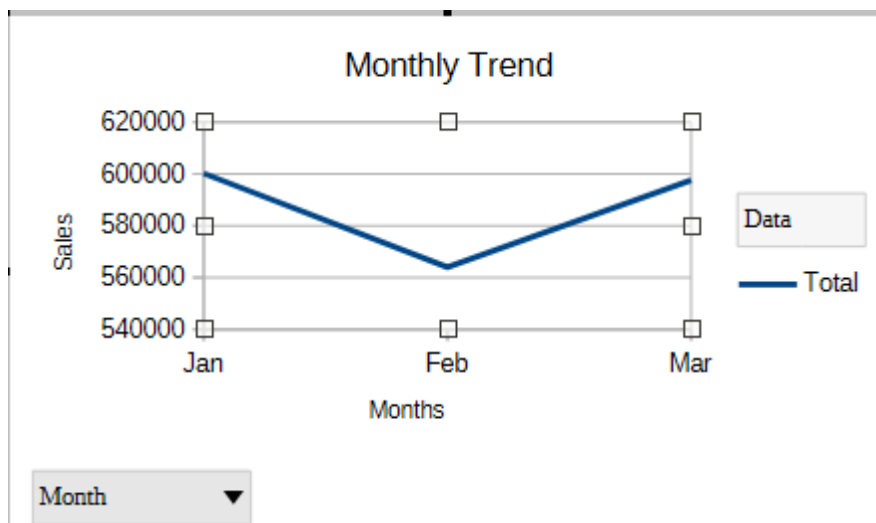
1.Region-wise Sales(Using Bar Chart) :



Insights :

- The **East region** has the **highest sales (~500,000)**, making it the top-performing region.
- The **North region** follows closely with around **480,000**, showing strong performance.
- The **West region** has moderate sales (~420,000).
- The **South region** records the **lowest sales (~350,000)**.

2. Monthly Trend(Using Line Chart) :



Insights :

1. Upward Growth Trend

- Sales show a **clear increase from February to January**
- Indicates **strong business growth and improving performance**

2. Peak Performance in January

- January has the highest sales
- Suggests peak demand or successful strategies during this period

Task 123 : Identify inconsistent/anomalous data and report findings.

Objective:

The objective of this task is to **detect, analyze, and report irregularities or anomalies in the dataset** that may affect data accuracy and decision-making.

Date	Region	Product Category	Sales	Quantity	Month
2023-01-01	North	Groceries	7920	5	Jan
2023-01-01	East	Clothing	4000	2	Jan
2023-01-01	North	Clothing	276	1	Jan
2023-01-02	East	Clothing	499	1	Jan
2023-01-02	South	Electronics	2547	3	Jan
2023-01-02	North	Electronics	5008	8	Jan
2023-01-02	South	Groceries	1966	2	Jan
2023-01-02	East	Electronics	8910	6	Jan
2023-01-02	East	Electronics	1788	6	Jan
2023-01-02	West	Groceries	4390	10	Jan
2023-01-03	East	Groceries	16460	10	Jan
2023-01-03	West	Groceries	4392	3	Jan
2023-01-03	West	Clothing	8752	8	Jan
2023-01-03	North	Clothing	2799	3	Jan
2023-01-03	South	Groceries	5030	10	Jan
2023-01-03	South	Electronics	6188	7	Jan
2023-01-03	North	Furniture	1405	5	Jan
2023-01-03	South	Furniture	932	2	Jan
2023-01-04	West	Clothing	4150	5	Jan
2023-01-04	West	Electronics	9460	10	Jan
2023-01-04	East	Groceries	18150	10	Jan
2023-01-04	South	Electronics	4000	2	Jan
2023-01-05	East	Furniture	5283	9	Jan

File name : Real_World_Sales_Data.xlsx

Inconsistency Detection Results of the given file :

1. Missing Values :

- **No Missing Value Found**
- **Data is complete and well-filled**

2. Duplicate Records :

- **No duplicate rows**
- **Data is unique and clean**

3. Invalid Values :

No negative or zero values in:

- **Sales**
- **Quantity**

Data is logically valid

4. Conclusion

- **Overall dataset is clean and reliable**
- **Only few outliers need attention**

Task 124 : Recommend ways to improve performance.

Objective

The objective is to analyze the dataset to identify strengths, weaknesses, and patterns in sales performance, and to recommend data-driven strategies that enhance overall efficiency, increase revenue, optimize resource utilization, and support better business decision-making.

Performance Improvement Recommendations :

1. Focus on High-Performing Regions

Identify regions with highest sales (e.g., West/North)

Increase:

- **Marketing campaigns**
- **Product availability**
- **Expand successful strategies to low-performing regions**

Impact: Boost overall revenue

2. Optimize Low-Performing Regions

Regions with lower sales need attention

Actions:

- **Local promotions**
- **Better distribution/logistics**
- **Region-specific offers**

Impact: Balanced growth across regions

3. Promote High-Demand Product Categories

Identify top-selling categories (e.g., Electronics)

Increase:

- **Stock levels**
- **Advertising**
- **Bundle with slow-moving products**

Impact: Higher sales volume

Task 125: Use high-quality and detailed data in reports.

Objective

The objective is to use rich and comprehensive data in reporting to provide deeper insights, improve accuracy, and enhance the overall quality of analysis, enabling better understanding of trends, patterns, and relationships for effective decision-making.

1. Ensure Data Quality (Cleaning) :

a. Missing Values :

- **No Missing Value Found**
- **Data is complete and well-filled**

b. Duplicate Records :

- **No duplicate rows**
- **Data is unique and clean**

2. Add More Detailed Columns (Make Data “Rich”) :

- **Month → =TEXT(Date,"mmm")**
- **Year → =YEAR(Date)**
- **Revenue per Unit → =Sales/Quantity**

Result: More dimensions for analysis

F	G	H
Month	Year	<u>RPU</u>
Jan	2023	1584
Jan	2023	2000
Jan	2023	276
Jan	2023	499
Jan	2023	849
Jan	2023	626
Jan	2023	983
Jan	2023	1485
Jan	2023	298
Jan	2023	439
Jan	2023	1646
Jan	2023	1464
Jan	2023	1094
Jan	2023	933
Jan	2023	503
Jan	2023	884
Jan	2023	281
Jan	2023	466
Jan	2023	830
Jan	2023	946
Jan	2023	1815
Jan	2023	2000
Jan	2023	587

3. Use Pivot Tables for Deep Insights :

Analyze:

- Sales by Region + Product Category
- Monthly Sales Trends

Result: Multi-level insights

Task 126 : Evaluate report performance, consistency, and business relevance.

Objective:

The objective is to evaluate the report for performance, consistency, and business relevance, ensuring it is accurate, well-structured, and useful for decision-making.

Evaluation of Report Performance, Consistency, and Business Relevance :

In this task, I evaluated the overall quality of the report generated from the Real World Sales Data. First, I analyzed the report performance by checking whether the visualizations and summaries clearly represent the data trends such as sales growth, regional performance, and product category contribution. The report was reviewed to ensure it is easy to understand and supports quick decision-making.

Next, I verified the consistency of the report by ensuring that the data used across tables, charts, and summaries is uniform and accurate. I checked for matching totals, correct labeling, and consistent formatting in all visual elements like graphs and pivot tables.

Finally, I assessed the business relevance of the report by linking the insights to real-world business decisions. The analysis focused on identifying high-performing regions, top-selling product categories, and areas needing improvement. This ensures that the report is not only technically correct but also useful for strategic planning and performance improvement.

Overall, the report was validated for clarity, accuracy, consistency, and its usefulness in supporting business decisions.

Task 128 : Analysis of Possible Decision Outcomes .

Objective:

The objective is to analyze possible outcomes of business decisions using sales data, helping to predict results, reduce risks, and support better strategic planning.

Analysis of Possible Decision Outcomes :

In this task, I analyzed possible outcomes of business decisions using the Real World Sales Data along with the monthly trend chart. The trend shows that sales were highest in January, dropped significantly in February, and then recovered in March.

Based on this pattern, I evaluated decisions such as improving performance during low-sales periods (February), increasing focus on high-performing months, and adjusting strategies like promotions or inventory planning. The February decline indicates a potential risk area, while the March recovery suggests that corrective actions or increased demand can improve results.

By using charts and data analysis, I predicted how different decisions could impact future sales. This helps in reducing risks, improving weak areas, and making better business decisions for consistent growth.

Task 129 : Implement real-time data updates in reports.

Objective

To design a dynamic reporting system where data automatically updates when new entries are added, ensuring accurate, up-to-date insights and enabling faster, data-driven decision-making without manual intervention.

Step 1: Preparing Data

Date ▼	Region ▼	Product Category ▼	Sales ▼	Quantity ▼
2023-01-01	North	Groceries	7920	5
2023-01-01	East	Clothing	4000	2
2023-01-01	North	Clothing	276	1

In this step, I ensured that the dataset contains the required columns: Date, Region, Product Category, Sales, and Quantity. Since LibreOffice Calc does not have a direct “Table” feature like Excel, I prepared the data by selecting the entire dataset and applying AutoFilter using Data → AutoFilter.

Step 2: Create Pivot Tables

a) Sales by Region :

Region ▼	Sum - Sales
East	501428
North	481407
South	357687
West	421345
Total Result	1761867

b) Monthly Sales Trend

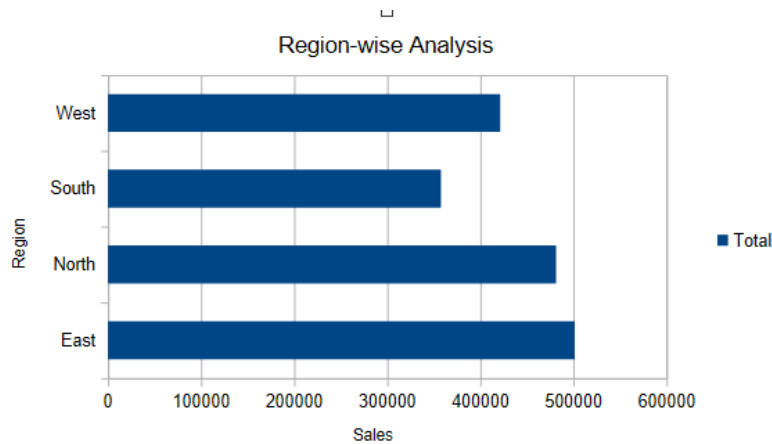
Month ▼	Sum - Sales
Jan	600179
Feb	564031
Mar	597657
Total Result	1761867

These will be the base for your dashboard charts

Step 3: Insert Charts :

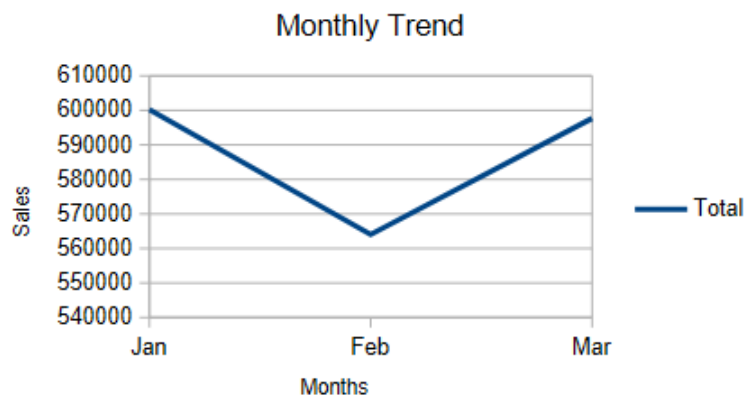
a) Region-wise Comparison :

- Insert Bar Chart → Region comparison



b) Monthly Trend :

- Insert Line Chart → Monthly trend



Charts will automatically update when data changes

Task 130 : Create a Report Using Spreadsheets .

Objective :

To organize, analyze, and present data effectively using spreadsheet tools by applying formulas, charts, and tables, enabling clear insights and supporting informed decision-making.

1. Converted the Sales in Currency format :

D	
Sales	
₹7,920.00	
₹4,000.00	
₹276.00	
₹499.00	

I organized and cleaned the dataset by applying proper formatting, including converting numerical values into currency format. I performed basic calculations such as total and average sales to understand overall performance.

I created pivot tables to summarize the data by key dimensions like region, product category, and time, which helped in identifying trends and comparisons. Additionally, I developed visualizations such as bar charts and line graphs to represent the data clearly.

Overall, I built a structured and interactive report that provides meaningful insights and supports effective decision-making.